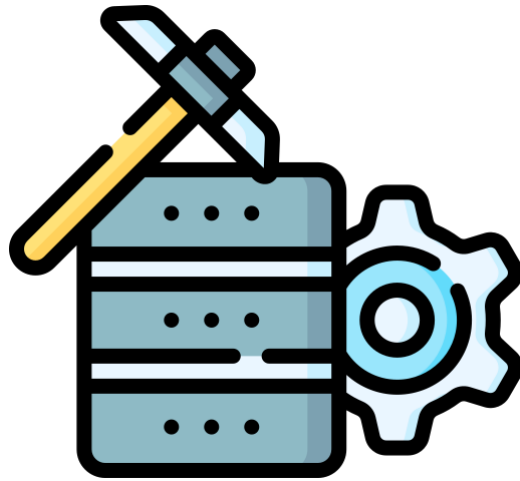




DATA MINING

2025

FINAL BANK

(Solved)



SENIOR
25

BY QAHWA TEAM

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Choose the correct answer:

1.	_____ is an essential process where intelligent methods are applied to extract data patterns .			
	a) Data warehousing	b) Text mining	c) Data mining	d) Data selection
2.	Data mining is best described as the process of _____.			
	a) identifying patterns in data	b) deducing relationships in data	c) representing data	d) simulating trends in data
3.	The various aspects of data mining methodologies is/are _____ I. Mining various and new kinds of knowledge. II. Mining knowledge in multidimensional space. III. Pattern evaluation and pattern or constraint-guided mining. IV. Handling uncertainty, noise, or incompleteness of data.			
	a) I, II, IV only	b) II, III, IV only	c) I, II, III only	d) I, II, III, IV
4.	the patterns that can be discovered from a given database can be of			
	a) Multiple type always	b) More than one type	c) One type only	d) On specific type
5.	If the mean < median the data is _____ skewed.			
	a) positively	b) negatively	c) symmetric	d) correlated
6.	If the mean is less than median the data is _____			
	a) Positively skewed	b) Negatively skewed	c) symmetric	d) correlated
7.	If the mean is smaller than the median then this might be an indication that the data is _____			
	a) positively skewed	b) negatively skewed	c) symmetric	d) correlated
8.	If the mean > median the data is _____ skewed.			
	a) positively	b) negatively	c) symmetric	d) correlated
9.	If the mean is more than median the data is _____			
	a) Positively skewed	b) Negatively skewed	c) symmetric	d) correlated
10.	If the mean is larger than the median then this might be an indication that the data is _____			
	a) positively skewed	b) symmetric	c) correlated	d) negatively skewed
11.	If the mean = median the data is _____ skewed.			
	a) positively	b) negatively	c) symmetric	d) correlated
12.	If the mean is equal to median the data is _____			
	a) Positively skewed	b) Negatively skewed	c) symmetric	d) correlated
13.	If the mean, median and mode of a distribution are 5, 6, 7 respectively, then the distribution is: (mean < median, 5 < 6, thus it is negatively skewed)			
	a) skewed negatively	b) not skewed	c) symmetrical	d) bimodal
14.	Nonparametric data reduction strategies include all the following except _____			
	a) Histograms	b) Regression	c) Clustering	d) Sampling
15.	Parametric data reduction strategies include _____			
	a) Histograms	b) Regression	c) Clustering	d) Sampling



16.	If you want to <i>give all attributes an equal weight</i> , which preprocess task will you use_____			
	a) Cleaning	b) Integration	c) Transformation	d) Reduction
17.	Task of inferring a model from labeled training data is _____			
	a) Clustering analysis	b) Association rule	c) Transformation	d) Classification
18.	Regression is a method of all the following except _____			
	a) Cleaning	b) Integration	c) Transformation	d) Reduction
19.	The bottleneck of the Apriori algorithm is caused by _____			
	a) No. of rules	b) Min. confidence	c) No. of candidate	d) Correlation
20.	The bottleneck of the Apriori algorithm is caused by all the following except			
	a) the number of association rules	b) the number of scans required	c) the computations of support for candidates	d) the number of generated candidates
21.	What is the mean of the following numbers? 10,39,71,42,39,76,38,25 $\text{mean} = \frac{\text{sum of elements}}{\text{number of elements}} = \frac{10 + 39 + 71 + 42 + 39 + 76 + 38 + 25}{8} = 42.5$			
	a) 42	b) 39	c) 42.5	d) 35
22.	The problem that related to calculate the mean is _____			
	a) sensitivity to outlier	b) time consuming	c) noise	d) missing data
23.	The problem that related to calculating the mean is _____			
	a) Noise	b) Missing data	c) Time consuming	d) Sensitivity to outlier
24.	If all the proper subsets of an itemset are frequent , then the itemset itself _____			
	a) must also be frequent	b) may be frequent	c) Un frequent	d) is strong
25.	Based on Apriori property, all nonempty subsets of frequent itemset _____.			
	a) must also be frequent	b) may be frequent	c) Un frequent	d) is strong
26.	If an itemset is frequent , then all the proper subsets of it _____			
	a) May be frequent	b) Is strong	c) Must be infrequent	d) Must be frequent
27.	If any of the proper subsets of an itemset are not frequent , then the itemset itself _____			
	a) Is weak	b) Must be infrequent	c) My be frequent	d) Must also be frequent



28.	If the <u>lift measure</u> of the items bread and rice is equal to 1.5 this means bread and rice are _____ (more than 1 → positive)			
	a) positive correlated	b) negative correlated	c) independent	d) uncorrelated
29.	if the value of the <u>lift measure</u> of A and C = 1.5 then the item A and C are _____ (more than 1 → positive)			
	a) positive correlated	b) negative correlated	c) independent	d) frequent
30.	If the <u>lift measure</u> of the items bread and rice is equal to 0 this means bread and rice are _____ (less than 1 → negative)			
	a) positive correlated	b) negative correlated	c) independent	d) uncorrelated
31.	If the lift measure of the items bread and rice is equal to -4 this means bread and rice are _____ (less than 1 → negative)			
	a) positive correlated	b) negative correlated	c) independent	d) uncorrelated
32.	If the <u>lift</u> measure of items bread and rice if equal 0.5 this means that (less than 1 → negative)			
	a) if client buy bread they are more likely to but rice	b) if client buy bread they are less likely to but rice	c) if client buy bread they can buy rice or not with the same probability	d) none of them
33.	If the <u>lift measure</u> of the items bread and rice is equal to 1 this means bread and rice are _____ (equal to 1 → independent)			
	a) positive correlated	b) negative correlated	c) independent	d) None
34.	if the value of the <u>lift=1</u> for item A and C, the item A and C are _____ (equal to 1 → independent)			
	a) positive correlated	b) negative correlated	c) independent	d) frequent
35.	If the value of chi-square test < table value this means the two are _____			
	a) positive correlated	b) negative correlated	c) independent	d) uncorrelated
36.	If the value of chi-square test > table value this means the two are _____			
	a) positive correlated	b) negative correlated	c) independent	d) frequent
37.	If the support of the rule > the min_support this means the rule is _____			
	a) good	b) frequent	c) not frequent	d) strong
38.	If the support of the rule < the min_support this means the rule is _____			
	a) good	b) frequent	c) not frequent	d) strong
39.	If the support of the rule > the min_support & confidence of the rule > the min_confidence this means the rule is _____			
	a) good	b) frequent	c) correlated	d) strong
40.	If the confidence of the rule < the min_confidence this means the rule is _____			
	a) good	b) frequent	c) not strong	d) strong
41.	The full form of KDD is _____			
	a) Knowledge Database Design	b) Knowledge Discovery in Database	c) Knowledge Data Definition	d) Kind Data Distance



42.	The KDD process consists of _____ steps.			
	a) three	b) four	c) five	d) six
43.	The output of KDD is _____.			
	a) Data	b) Query	c) Knowledge	d) Useful information
44.	The output of KDD is _____			
	a) Data	b) Information	c) Query	d) Useful information
45.	Various visualization techniques are used in _____ step of KDD			
	a) interpretation	b) data mining	c) transformation	d) selection
46.	This approach is best when we are interested in finding all possible interactions among a set of attributes			
	a) decision tree	b) association rules	c) K-Means algorithm	d) genetic learning
47.	_____ replaces original data volume by smaller data representation .			
	a) Dimensionality reduction	b) Numerosity reduction	c) Compression	d) Transformation
48.	Reducing the number of attributes to solve the high dimensionality problem is called as _____			
	a) Cleaning	b) Overfitting	c) Dimensionality	d) dimensional reduction
49.	Reducing the number of attributes to solve the high dimensional problem is called as _____			
	a) Overfitting	b) Cleaning	c) Dimensionality	d) Dimensionality reduction
50.	This technique uses <u>mean and standard deviation scores</u> to transform real-valued attributes _____			
	a) z-score normalization	b) min-max normalization	c) decimal scaling	d) logarithmic normalization
51.	This data transformation technique works when well when minimum and maximum value for a real-valued attribute are known			
	a) Logarithmic normalization	b) Decimal scaling	c) Min-max normalization	d) z-score normalization
52.	_____ is the result of a tuple firing more than one rule with different class predictions			
	a) interested rule	b) Strong rule	c) Rule conflict	d) Association rule
53.	if rule conflict occurs , you will choose the rule with _____			
	a) largest antecedent	b) smallest antecedent	c) Both of them	d) None of them
54.	Data used to build a data mining model is _____			
	a) validation data	b) training data	c) test data	d) hidden data
55.	Data used to evaluate a data mining model is _____			
	a) validation data	b) training data	c) test data	d) hidden data
56.	The confidence for the association rule $(X) \rightarrow (Y)$ was determined to be 0.90. What does the value 0.90 mean?			
	a) 90% of customers who buy X buy also Y	b) 90% of customers buy Y and X	c) 90% of customers who buy X	d) 90% of customers who buy Y



57.	The support for the association rule {bread} → (milk) was determined to be 0.95. What does the value 0.95 mean?							
	a)	95% of customers who buy bread buy also milk	b)	95% of customers buy bread and milk	c)	95% of customers who buy milk buy also bread	d)	95% of customers who buy bread don't buy milk
58.	The confidence for the association rule {Bread} → {Milk} was determined to be 0.95. what does the value 0.95 mean?							
	a)	95% of customers who buy milk buy also bread						
	b)	95% of customers who buy bread don't like milk						
	c)	95% of customers who buy bread buy also milk						
59.	The correlation analysis is used to _____							
	a)	extract rules	b)	define support values	c)	eliminate misleading rules	d)	remove the noise
60.	The correlation coefficient for two real-valued attributes is -0.85 . What does this value tell you?							
	a)	The attributes are not linearly related						
	b)	The attributes show a curvilinear relationship						
	c)	As the value of one attribute decreases the value of the second attribute increases						
61.	The correlation coefficient for two real-valued attributes is 0.95 , what does this value tell you?							
	a)	As the value of one attribute increases the value of the second attribute also increases						
	b)	As the value of one attribute decreases the value of the second attribute increases						
	c)	The attributes are not linearly related						
62.	Identify the outlier for the given data? 23, 34, 27, <u>7</u> , 30, 26, 28, 31, 34							
	a)	7	b)	23	c)	31	d)	34
	What does the term 'outlier' mean?							
	a)	A score that is left out of the analysis because of missing data						
63.	b)	The arithmetic mean						
	c)	A type of variable that cannot be quantified						
	d)	An extreme value at either end of a distribution						
	Which statement about outliers is true?							
64.	a)	It should be part of the training database but should not be present in the test data						
	b)	It should be identified and removed from a dataset						
	c)	The nature of the problem determines how outliers are used						
	d)	It should be part of the best dataset but should not be present in the training data						



65.	Which of the following lists all parts of the five-number summary ?						
	a)	Mean, Median, Mode, Range and Total					
	b)	Minimum, Quartile1, Median, Quartile3 and Maximum					
	c)	Minimum, Maximum, Range, Mean and Median					
	d)	Smallest, Q1, Q2, Q3, Biggest					
66.	Which of the following lists all parts of the five-number summary ?						
	a)	Mean, Median, Mode, Range, and Total					
	b)	Minimum, Q1, Q2, Q3, and Maximum					
	c)	Minimum, Maximum, Range, Mean, and Median					
	d)	Smallest, Q1, Q2, Q3, Biggest					
67.	The probability that two items are bought is known as _____						
	a)	Confidence	b)	Correlation	c)	Support	d)
68.	Which of the following are interestingness measures for association rules?						
	a)	lift	b)	compactness	c)	recall	d)
69.	The terminating conditions in decision tree include all the following except _____						
	a)	No tuples for a given branch	b)	No remaining attributes	c)	All tuples belong to the same class	d)
70.	If you use decision tree induction for prediction and no rules are triggered , then you will use _____.						
	a)	Fallback rule	b)	Rule with largest antecedent	c)	Rule with highest quality	d)
71.	In the decision tree induction, you will choose the attribute with _____ information gain to be a splitting attribute						
	a)	large	b)	small	c)	positive	d)
72.	If the information gain of age , income and gender attributes are 0.42 , 0.24 and 0.024 which one will you choose as splitting attribute (LARGEST information gain)						
	a)	age	b)	income	c)	gender	d)
73.	If the information gain of age, income, department, and job attributes are 0.24, 0.204, 0.32 and 0.023 respectively which one you will choose as the splitting attribute (LARGEST information gain)						
	a)	age	b)	income	c)	department	d)
74.	In the decision tree induction, the leaf is _____						
	a)	A question	b)	An attribute value	c)	A class label	d)
75.	In the decision tree induction, the branch is _____						
	a)	A test outcome	b)	An attribute value	c)	A class label	d)
76.	In the decision tree induction, the internal node is _____						
	a)	A test outcome	b)	A test on an attribute	c)	A class label	d)



77.	_____ is the correlation analysis measure for nominal attribute							
	a)	lift	b)	covariance	c)	Correlation coefficient	d)	Chi-square
78.	_____ is the correlation analysis measure for numeric attributes							
	a)	lift	b)	covariance	c)	Correlation coefficient	d)	Chi-square
79.	_____ measure how strongly one attribute implies the other, based on the available data							
	a)	lift	b)	covariance	c)	correlation analysis	d)	Chi-square
80.	The SD spread out over a large range of values, this means that the data has _____							
	a)	low SD	b)	high SD	c)	positively skewed	d)	negatively skewed
81.	The SD observations tend to be very close to the mean , this means that the data has _____							
	a)	low SD	b)	high SD	c)	positively skewed	d)	negatively skewed
82.	For the following association rule: Computer → Webcam (60%, 100%): Which of the following is true? I. 100% of costumers bought both a computer and a webcam II. 60% of costumers bought both a computer <u>and</u> a webcam III. 100% of costumers who bought a computer bought <u>also</u> a webcam IV. 60% of costumers who bought a computer bought also a webcam							
	a)	II only	b)	III Only	c)	I and IV	d)	II and III
83.	We have Market Basket data for 1,000 rental transactions at a Video Store. There are four videos for rent -- Video A, Video B, Video C and Video D. The probability that both Video C and Video D are rented at the same time is known as _____							
	a)	Correlation	b)	support	c)	lift	d)	confidence
84.	A data _____ represents an entity							
	a)	object	b)	attribute	c)	value	d)	none
85.	An _____ is a data field , representing a characteristic or feature of a data object							
	a)	object	b)	attribute	c)	value	d)	none
86.	In a binary attribute, _____ both states are equally valuable and have the same weight							
	a)	Symmetric	b)	Asymmetric	c)	Qualitative	d)	data
87.	In a binary attribute, _____ states are not equally important							
	a)	Symmetric	b)	Asymmetric	c)	Qualitative	d)	data
88.	In a binary attribute, if one state is more valuable and has high weight than the other , it is called _____.							
	a)	Symmetric	b)	Asymmetric	c)	discreet	d)	None
89.	A finite or <u>countably infinite</u> set of values, integers or otherwise of attribute you will use _____							
	a)	interval-scaled	b)	ratio-scaled	c)	discrete	d)	nominal
90.	A measured <u>on a scale of equal-size units</u> of attribute you will use _____							
	a)	interval-scaled	b)	ratio-scaled	c)	discrete	d)	nominal
91.	What kind of attribute will you use to define the temperature values ?							
	a)	interval-scaled	b)	ratio-scaled	c)	discrete	d)	nominal



92.	If you need to define the <u>temperature attribute</u> , then what kind of attribute you will use?			
	a) interval-scaled	b) ratio-scaled	c) discrete	d) nominal
93.	If a <u>person's height</u> is measured in <u>inches</u> then what kind of attribute you will use_____			
	a) interval-scaled	b) ratio-scaled	c) discrete	d) nominal
94.	The attribute data type of <u>number of telephones in your house</u> is _____.			
	a) Nominal	b) Ordinal	c) Interval	d) Ratio
95.	What kind of attribute will you use to define the telephone number ?			
	a) Nominal	b) discrete	c) interval scaled	d) ratio scaled
96.	Which type of attribute can you find the median?			
	a) Numeric	b) Ordinal	c) None of them	d) Both of them
97.	Bootstrapping allows us to _____.			
	a) Choose the same training instance several times			
	b) Build models with alternative subsets of the training data several times			
	c) Choose the same test set instance several times			
	d) Test a model with alternative subsets of the test data several times			
98.	This step of the KDD process model deals with noisy data			
	a) Data transformation	b) Creating a target dataset	c) Data preprocessing	d) Data mining
99.	_____ fill in missing values, smooth out noise while identifying outliers , and correct inconsistencies in the data			
	a) Data cleaning	b) Data integration	c) Data reduction	d) Discretization
100.	Which of the following is the process of <u>detecting and correcting the wrong data</u> ?			
	a) Data cleaning	b) Data integration	c) Data reduction	d) Discretization
101.	_____ filling in missing values, smoothing noisy data , identifying or removing outliers , and resolving inconsistencies			
	a) Data cleaning	b) Data integration	c) Data reduction	d) Discretization
102.	Removing noise and inconsistent data refers to which phase of the knowledge discovery process?			
	a) Data cleaning	b) Data integration	c) Data reduction	d) Discretization
103.	_____ <u>include</u> data from multiple sources in your analysis, map semantic concepts, infer attributes			
	a) Data cleaning	b) Data integration	c) Data reduction	d) Discretization
104.	Which of the following is the process of <u>combining</u> data from different sources ?			
	a) Data cleaning	b) Data integration	c) Data selection	d) Data transformation
105.	_____ obtain a reduced representation of the data set that is much smaller in volume, while producing almost the same analytical results			
	a) Data cleaning	b) Data integration	c) Data reduction	d) Discretization
106.	_____ raw data values for attributes are replaced by ranges or higher conceptual levels			
	a) Data cleaning	b) Data integration	c) Data reduction	d) Discretization



107.	Concept Hierarchy is used to _____.							
	a)	Data cleaning	b)	Data integration	c)	Data reduction	d)	Data Discretization
108.	Which of the following is not a data mining functionality?							
	a)	Characterization and Discrimination	b)	Classification and regression	c)	Selection and interpretation	d)	Clustering and Analysis
109.	_____ is a comparison of the general features of the target class data objects against the general features of objects from one or multiple contrasting classes.							
	a)	Data Characterization	b)	Data Classification	c)	Data discrimination	d)	Data selection
110.	_____ is the process of <i>finding a model that describes and distinguishes data classes</i> or concepts.							
	a)	Data Classification	b)	Data Characterization	c)	Data integration	d)	Data cleaning
111.	_____ is the process of finding a model that describes and distinguishes data classes							
	a)	Data characterization	b)	Data classification	c)	Data discrimination	d)	Data selection
112.	_____ is the summarization of the general characteristics or features of a target class of data.							
	a)	Data Classification	b)	Data Characterization	c)	Data integration	d)	Data cleaning
113.	Support and Confidence are measures of _____							
	a)	rule interestingness	b)	association rule	c)	Applications	d)	Motivation
114.	The measures that are used to determine the degree of extent of variation in a dataset are called _____.							
	a)	median	b)	mean	c)	measures of dispersion	d)	measures of central tendency
115.	In this association rule : Computer → antivirus_SW [2%, 60%], Support = _____							
	a)	2%	b)	8%	c)	60%	d)	75%
116.	In this association rule : Computer → antivirus_SW [2%, 60%], Confidence = _____							
	a)	2%	b)	8%	c)	60%	d)	75%
117.	In FP-tree, when you go down the <i>frequency</i> _____.							
	a)	increases	b)	improves	c)	is fixed	d)	decreases
118.	Supervised learning differs from unsupervised clustering in that <u>supervised learning requires</u>							
	a)	at least one input attribute.	b)	input attributes to be categorical	c)	at least one output attribute	d)	output attributes to be categorical
119.	Supervised learning and unsupervised clustering <u>both require at least one</u> _____.							
	a)	hidden attribute	b)	output attribute	c)	input attribute	d)	categorical attribute



120.	The correlation between the number of years an employee has worked for a company and the salary of the employee is 0.75. What can be said about employee salary and years worked?							
	a)	There is no relationship between salary and years worked.						
	b)	Individuals that have worked for the company the longest have higher salaries						
	c)	Individuals that have worked for the company the longest have lower salaries						
	d)	The majority of employees have been with the company a long time						
121.	Unlike traditional production rules, association rules_____							
	a)	allow the same variable to be an input attribute in one rule and an output attribute in another rule						
	b)	allow more than one input attribute in a single rule.						
	c)	require input attributes to take on numeric values						
	d)	require each rule to have exactly one categorical output attribute.						
122.	Given desired class C and population P, lift is defined as_____							
	a)	the probability of class C given population P divided by the probability of C given a sample taken from the population.						
	b)	the probability of population P given a sample taken from P						
	c)	the probability of class C given a sample taken from population P.						
	d)	the probability of class C given a sample taken from population P divided by the probability of C within the entire population P						
123.	Association rule support is defined as _____							
	a)	the percentage of instances that contain the antecedent conditional items listed in the association rule						
	b)	the percentage of instances that contain the consequent conditions listed in the association rule						
	c)	the percentage of instances that contain all items listed in the association rule						
	d)	the percentage of instances in the database that contain at least one of the antecedent conditional items listed in the association rule						
124.	Combining classifiers improves _____.							
	a)	interpretability	b)	accuracy	c)	Similarity	d)	correlations
125.	Point out the correct statement:							
	a)	Combining classifiers reduces accuracy						
	b)	Combining classifiers improves interpretability						
	c)	Combining classifiers improves Accuracy						
	d)	None of them						
126.	Point out the correct statement							
	a)	K-means algorithm is used for classification						
	b)	K-nearest neighbor is the same as K-means						
	c)	K-means algorithm aims to partition n objects into k subgroups						
	d)	None						



127.	The <i>K-Means algorithm terminates</i> when _____.									
	a)	A user defined minimum value for the summation of squared error differences between instances and their corresponding cluster center is seen								
	b)	The cluster centers for the current iteration are identical to the cluster centers for the previous iteration.								
	c)	The number of instances in each cluster for the current iteration is identical to the number of instances in each cluster of the previous iteration								
	d)	The number of clusters formed for the current iteration is identical to the number of clusters formed in the previous iteration								
128.	Which of the following best describes the process of finding the <u>interquartile range</u> for a set of data?									
	a)	ADD the biggest and smallest numbers	b)	Place the number in order from least to greatest then find the middle	c)	Find the difference between the Maximum and the Minimum	d)	Subtract Q3 from Q1		
129.	What is the term for the median of the <u>lower half of the data</u> ?									
	a)	Lower Quartile	b)	Upper Quartile	c)	Median	d)	Maximum		
130.	What is the term that means the <u>middle data value</u> ?									
	a)	Mean	b)	Median	c)	Mode	d)	Range		
131.	What is the mode?									
	a)	# happening the most	b)	the average	c)	biggest - smallest	d)	the middle #		
132.	The heights of some students are given. <table><tr><td>158 cm</td><td>172 cm</td><td>164 cm</td></tr><tr><td>164 cm</td><td>167 cm</td><td>159 cm</td></tr></table> What is the <u>range</u> of the heights? (<u>range = max – min</u>)				158 cm	172 cm	164 cm	164 cm	167 cm	159 cm
158 cm	172 cm	164 cm								
164 cm	167 cm	159 cm								
	a)	13cm	b)	14cm	c)	164cm	d)	330cm		
133.	Supervised learning and unsupervised clustering both require at least one _____.									
	a)	hidden attribute	b)	output attribute	c)	input attribute	d)	categorical attribute		
134.	Data mining can also applied to other forms such as _____ i) Data streams ii) Sequence data iii) Networked data iv) Text data v) Spatial data									
	a)	i, ii, iii and v only	b)	ii, iii, iv and v only	c)	i, iii, iv and v only	d)	All i, ii, iii, iv and v		
135.	Strategic value of data mining is _____									
	a)	cost-sensitive	b)	work-sensitive	c)	time-sensitive	d)	technical-sensitive		



136.	Bayesian classifiers is _____						
a)	A class of learning algorithm that tries to find an <u>optimum classification</u> of a set of examples using the probabilistic theory.						
b)	Any mechanism employed by a learning system to constrain the search space of a hypothesis.						
c)	An approach to the design of learning algorithms that is inspired by the fact that when people encounter new situations, they often explain them by reference to familiar experiences, adapting the explanations to fit the new situation						
d)	None of them						
137.	Classification is _____						
a)	A subdivision of a set of examples into a number of classes						
b)	A measure of the accuracy, of the classification of a concept that is given by a certain theory.						
c)	The task of assigning a classification to a set of examples						
d)	None of them						
138.	Which of the following measures of central tendency tends to be most influenced by an <u>extreme score</u> ?						
a)	median	b)	mode	c)	mean	d)	Standard deviation
139.	Which of the following is <u>not</u> a measure of central tendency?						
a)	mean	b)	standard deviation	c)	median	d)	mode
140.	In a group of 12 scores, the largest score is <u>increased</u> by 36 points. What effect will this have on the mean of the scores ? ($36 / 12 = 3$)						
a)	it will be increased by 12 points	b)	it will remain unchanged	c)	it will be increased by 3 points	d)	it will increase by 36 points
141.	All the following are considered as characteristic of the interesting patterns to be discovered through a data mining process except _____.						
a)	no-trivial	b)	explicit	c)	previously unknown	d)	useful
142.	Which of the following <u>clustering</u> requires merging approach ?						
a)	Partitional	b)	Hierarchical	c)	Density-based	d)	Grid-based
143.	Which of the following clustering has fast processing time ?						
a)	Hierarchical	b)	Grid-based	c)	Density-based	d)	Partitional
144.	Point out the wrong statement:						
a)	K-means clustering is density based method						
b)	K-means clustering aims to partition n observations into k clusters						
c)	K-nearest neighbor is same as k-means						
d)	None						
145.	What classifiers are normally considered to be <u>easy to interpret</u> ?						
a)	SVM	b)	Naive Bayes	c)	Decision trees	d)	K-Nearest Neighbor



146.	What is the Naive Bayes Algorithm used for?			
	a)	Generate mining models		
	b)	Estimating the probability of a class value during classification and prediction		
	c)	To make decisions for reporting		
	d)	Both A & B		
147.	Selecting data so as to assure that <i>each class is properly represented</i> in both the training and test set.			
	a)	cross validation	b)	stratification
	c)	verification	d)	bootstrapping
148.	Specificity is _____			
	a)	TN / N	b)	TP / P
	c)	FP + FN / P + N	d)	TP + TN / P + N
149.	Sensitivity is _____			
	a)	TN / N	b)	TP / P
	c)	FP + FN / P + N	d)	TP + TN / P + N
150.	Error rate is _____			
	a)	IN / N	b)	TP / P
	c)	FP + FN / P + N	d)	TP + TN / P + N
151.	Accuracy is _____			
	a)	IN / N	b)	TP / P
	c)	FP + FN / P + N	d)	TP + TN / P + N
152.	RANDOMLY allocate 2/3 of data for training and remaining 1/3 for testing is called _____.			
	a)	Holdout	b)	k-fold cross-validation
	c)	Stratified k-fold cross-validation	d)	Random subsampling
153.	_____ repeat holdout k times and take average accuracy			
	a)	Holdout	b)	k-fold cross-validation
	c)	Stratified k-fold cross-validation	d)	Random subsampling
154.	The Pruning make the decision tree more _____			
	a)	complex	b)	simple
	c)	accuracy	d)	correlated
155.	if the object deviates significantly from the rest of the dataset, it will be _____			
	a)	global outlier	b)	local outlier
	c)	contextual outlier	d)	collective outlier
156.	Intrusion Detection can be considered as a case of _____.			
	a)	global outlier	b)	local outlier
	c)	contextual outlier	d)	collective outlier
157.	if the object deviates in a specific context , it will be _____			
	a)	global outlier	b)	local outlier
	c)	contextual outlier	d)	collective outlier
158.	If we need to find the Deviate values of a Temperature attribute, we need to deal with it as _____.			
	a)	global outlier	b)	local outlier
	c)	contextual outlier	d)	collective outlier
159.	_____ is s subset of objects collectively deviates significantly from the dataset			
	a)	Global outlier	b)	local outlier
	c)	contextual outlier	d)	collective outlier
160.	DoS attacks can be considered as a case of _____.			
	a)	global outlier	b)	local outlier
	c)	contextual outlier	d)	collective outlier



161.	A core point in DBSCAN is an object whose e-neighborhood contains _____ object			
	a) Null	b) at least MinPts	c) MinPts	d) at most MinPts
162.	What is the meaning of the previous example?			
	a) The cluster containing o is impact and o is also impact with other cluster			
	b) The cluster containing o is impact and o is far from with other cluster			
	c) Both a and b			
	d) None of them			
163.	An object p is directly _____ from another object q if and only if a core object and p is in the e-neighborhood of q.			
	a) a come	b) exclude	c) density-connected	d) density-reachable
164.	Clustering algorithm which can find clusters of arbitrary shape			
	a) Agglomerative	b) Divisive	c) DBSCAN	d) K-means
165.	_____ clustering algorithm initially assumes that all data instance in a single large cluster			
	a) Divisive	b) conceptual	c) K-Means	d) agglomerative
166.	if you use unsupervised algorithm, you will use _____			
	a) K-Means	b) Decision tree	c) FP-Growth	d) K-nearest neighbor
167.	In K-nearest neighbor algorithm K stands for _____ while K-means algorithm K stand for _____.			
	a) no. of cluster, no. of neighbors	b) no. of neighbors, no. of clusters	c) no. of rules, no. of classes	d) no. of classes, no. of rules
168.	Clustering algorithms which can find clusters of spherical shape are _____			
	a) K-mean	b) DBSCAN	c) KNN	d) Agglomerative
169.	clustering algorithm initially select clusters centers randomly			
	a) Divisive	b) conceptual	c) K-Means	d) agglomerative
170.	Task of inferring a model from labelled training data is called _____			
	a) Classification	b) Association rule	c) Cluster analysis	d) Transformation
171.	In lazy learner we interest in the _____ distance			
	a) Equal	b) Different	c) Largest	d) Smallest
172.	If K-means finds clusters with very different sizes and with concave shapes this mean _____			
	a) It find good clusters	b) It fails	c) It needs more time	d) None of them
173.	Outliers _____ cluster			
	a) Belong to small cluster	b) Don't belong to any	c) Belong to sparse	d) All of them
174.	Which of the following is required by K-means clustering?			
	a) Initial guess as to cluster centroids	b) Number of clusters	c) Defined distance metric	d) All of them
175.	_____ is a way to model a relationship between two sets of variables as an equation?			
	a) Clustering	b) Classification	c) Regression	d) Association rule



176.	Which statement is true about the K-means algorithm ?			
	a)	All attributes values must be categorical		
	b)	The output attribute must be categorical		
	c)	All attributes must be numeric		
	d)	Attribute values may be either categorical or numeric		
177.	which of the following is finally produced by <u>hierarchical clustering</u> ?			
	a)	final estimate of cluster centroids	b)	tree showing how close thing are to each other
	c)	assignment of each point to clusters	d)	All of them
178.	Which of the following <u>clustering</u> requires merging approach ?			
	a)	Partitional	b)	Hierarchical
	c)	Density-based	d)	Grid-based
179.	Which of the following clustering has fast processing time ?			
	a)	Hierarchical	b)	Grid-based
	c)	Density-based	d)	Partitional
180.	_____ algorithm is bottom-up composition and start from each object has its own cluster.			
	a)	K-mean	b)	Agglomerative
	c)	Divisive	d)	B & C
181.	the clustering algorithm initially assumes that each data instance represents a single cluster			
	a)	Expectation maximization	b)	Agglomerative clustering
	c)	K-means clustering	d)	Conceptual clustering
182.	_____ partitioning a set of data object into clusters			
	a)	Classification	b)	Clustering
	c)	Association rule	d)	Both b and c
183.	in _____ the <u>groups</u> are not predefined			
	a)	prediction	b)	Clustering
	c)	Association rule	d)	summarization
184.	The process of <u>grouping</u> a set of objects into classes of similar objects is called _____			
	a)	prediction	b)	Clustering
	c)	Association rule	d)	summarization
185.	If you need to make a prediction you will use _____			
	a)	DBSCAN	b)	FP- Growth
	c)	K-nearest neighbor	d)	K-means
186.	<u>Remove sub-trees</u> from already constructed trees is a _____			
	a)	Remove noise	b)	Pre-pruning
	c)	Remove outlier	d)	Post-pruning
187.	_____ methods measure how well the clusters are separated.			
	a)	Silhouette coefficient	b)	Extrinsic
	c)	Intrinsic	d)	None of them
188.	If a customer is spending more than expected , the customer's intrinsic value is _____ their actual value.			
	a)	greater than	b)	less than
	c)	less than or equal to	d)	equal to
189.	_____ methods evaluate clustering by comparing clustering against ground truth			
	a)	Silhouette coefficient	b)	Extrinsic
	c)	Intrinsic	d)	None of them



190.	The silhouette coefficient is a method used to determine _____.							
	a)	the number of clusters	b)	the goodness of clusters	c)	the distribution of clusters	d)	the density of clusters
191.	Data Analysis can be done on _____.							
	a)	Structured data	b)	Semi-structured data	c)	Unstructured data	d)	All of them
192.	_____ shows <u>numerical comparisons between categories</u> . No. columns should not be too large .							
	a)	Column Chart	b)	Bar Chart	c)	Line Chart	d)	Scatter Plot
193.	_____ number of bars can be relatively large							
	a)	Column Chart	b)	Bar Chart	c)	Line Chart	d)	Scatter Plot
194.	_____ show the <u>change of data</u> over a continuous time interval or time span							
	a)	Column Chart	b)	Bar Chart	c)	Line Chart	d)	Scatter Plot
195.	_____ The filling of the color can <i>better highlight the trend information</i> .							
	a)	Column Chart	b)	Bar Chart	c)	Line Chart	d)	Area Chart
196.	_____ shows two variables in the form of <u>points</u> on a rectangular coordinate system.							
	a)	Column Chart	b)	Bar Chart	c)	Line Chart	d)	Scatter Plot
197.	_____ is a multivariate chart that is a variant of a scatter plot..							
	a)	Column Chart	b)	Bar Chart	c)	Bubble Chart	d)	Scatter Plot
198.	_____ display a series of steps and the completion rate for each step.							
	a)	Funnel Chart	b)	Pie Chart	c)	Line Chart	d)	Scatter Plot
199.	_____ represents one static number, divided into categories that constitute its individual portion.							
	a)	Column Chart	b)	Pie Chart	c)	Line Chart	d)	Scatter Plot
200.	_____ shows the timing of mission .							
	a)	Gantt Chart	b)	Bar Chart	c)	Line Chart	d)	Scatter Plot
201.	_____ is a visual representation of displaying a distribution of data , usually across groups, based on a five number summary .							
	a)	Column Chart	b)	Pie Chart	c)	Line Chart	d)	Box Plot
202.	_____ shows the relationship between two measures and provides rating information using varying colors or saturation.							
	a)	Column Chart	b)	Pie Chart	c)	Heatmap	d)	Scatter Plot
203.	_____ is a subject-oriented, integrated, time-variant, nonvolatile collection or data in support of management decisions.							
	a)	Data Mining	b)	Data Warehousing	c)	Document Mining	d)	Text Mining
204.	Data Warehouse stores _____.							
	a)	Historical data	b)	Up to date data	c)	Real-time data	d)	Statistical data



205.	A data warehouse is _____						
	a)	Read only					
	b)	Write only					
	c)	Read and write					
	d)	Execute only					
206.	A data warehouse is _____						
	a)	Contains numerous conventions and formats					
	b)	Updated by end user					
	c)	Organized around important subject areas					
	d)	Contain only current data					
207.	Business Intelligence and data warehousing is used for _____.						
	a)	Forecasting					
	b)	Data Mining					
	c)	Analysis of large volumes of product sales data					
	d)	All of the above					
208.	What is a common time horizon for data stored in a Data warehouse?						
	a)	Current year	b)	2-4 years	c)	5-6 years	d)
209.	Data warehouse contains _____ data this is <u>never found in the operational environment</u>						
	a)	Summary	b)	Informational	c)	Normalized	d)
210.	In what environment can data be updated?						
	a)	Data warehouse	b)	Data mining	c)	Operational	d)
211.	What is the main organizational justification for implementing a data warehouse?						
	a)	Cheaper ways of handling data movement					
	b)	Decision support					
	c)	Storing large volumes of data					
	d)	Providing access to data					
212.	An operational system is _____.						
	a)	used to run the business in real time and is based on historical data.					
	b)	used to run the business in real time and is based on current data.					
	c)	used to support decision making and is based on current data.					
	d)	used to support decision making and is based on historical data.					



213.	What are characteristics of an active data warehouse architecture?			
	a)	At least one data mart		
	b)	Data that has extracted from multiple internal and external		
	c)	Near real-time data updates		
	d)	All of the above		
214.	The data from the operational environment enter _____ of data warehouse.			
	a)	Current detail data	b)	Older detail data
	c)	Lightly summarized data	d)	Highly summarized data
215.	Which of the following is <u>not a component of a data warehouse</u> ?			
	a)	Metadata	b)	Current detail data
	c)	Lightly summarized data	d)	Component Key
216.	What term is used for descriptions of the data contained in the data warehouse?			
	a)	Metadata	b)	Operational data
	c)	Informational data	d)	Relational data
217.	Which of the following is <u>not a component of a data warehouse</u> ?			
	a)	Data extraction/cleaning/preparation programs		
	b)	Data warehouse data		
	c)	Data meta data		
218.	The data is stored, retrieved and updated in _____			
	a)	OLAP	b)	OLTP
	c)	SMTP	d)	FTP
219.	An system is market-oriented and is used for data analysis by knowledge workers, including managers, executives, and analysts.			
	a)	OLAP	b)	OLTP
	c)	Both of the above	d)	None of the above
220.	The core of the <u>multidimensional</u> model is the _____, which consists of a large set of facts and a number of dimensions .			
	a)	Multidimensional cube	b)	Dimensions cube
	c)	Data Cube	d)	Data model
221.	_____ are responsible for running queries and reports <u>against data warehouse tables</u> .			
	a)	Hardware	b)	Software
	c)	End users	d)	Middleware
222.	The type of relationship in star schema is _____			
	a)	Many to many	b)	One to one
	c)	One to many	d)	Many to one
223.	What type of relationships exist in a star schema between <u>dimension and the fact table</u> ?			
	a)	Many to many	b)	One to one
	c)	One to many	d)	Many to one
224.	Generally, a star schema is composed of _____ fact table(s)?			
	a)	One	b)	Two
	c)	Three	d)	Four
225.	_____ is a <u>good alternative to the star schema</u> .			
	a)	Star schema	b)	Snowflake schema
	c)	Fact constellation	d)	Star-snowflake schema



226.	What is detailed data in a fact table called?							
	a)	Homogeneous data	b)	Attribute data	c)	Atomic data	d)	Reporting data
227.	What do dimension tables commonly describe in a star schema?							
	a)	Descriptive entities	b)	Relevant facts	c)	Units of measures	d)	Data domains
228.	The biggest drawback of the <i>level indicator</i> in the classic star schema is that it <u>limits</u> _____.							
	a)	Flexibility	b)	Quantify	c)	Qualify	d)	Ability
229.	A <u>fact table in the middle connected to a set of dimension tables</u> is called _____.							
	a)	Star schema	b)	Snowflake schema	c)	Fact constellations	d)	None
230.	_____ is a refinement of star schema <u>where some dimensional hierarchy is normalized into a set of smaller dimension tables</u> , forming a shape similar to snowflake							
	a)	Star schema	b)	Snowflake schema	c)	Fact constellations	d)	None
231.	Multiple fact tables share dimension tables, viewed as a collection of stars , therefore called _____.							
	a)	Star schema	b)	Snowflake schema	c)	Fact constellations	d)	None
232.	The full form of OLAP is _____.							
	a)	Online Analytical Processing						
	b)	Online Advanced Processing						
	c)	Online Advanced Preparation						
	d)	Online Analytical Performance						
233.	_____ approach Starts with overall design and planning. (mature)							
	a)	Top-down	b)	Bottom-up	c)	Combined	d)	None
234.	_____ approach Starts with experiments and prototypes. (rapid)							
	a)	Top-down	b)	Bottom-up	c)	Combined	d)	None



True/False:

1.	Data mining is extraction of interesting (trivial , implicit, previously known and potentially useful) patterns or knowledge from huge amount of data (non-trivial, previously unknown)	F
2.	Association rules provide information in the form of "if-then" statements. (Decision tree)	F
3.	<u>Binning</u> is a method of reduction . (cleaning)	F
4.	Cluster is the process of finding a model that <u>describes and distinguishes data classes or concepts</u> . (Classification)	F
5.	<u>False Negatives</u> means <i>positive</i> tuples <i>incorrectly</i> labeled.	T
6.	<u>Intrinsic methods</u> measure how well the clusters are separated	T
7.	Mode can be found in the <u>nominal attribute</u> values.	T
8.	Recall is interestingness measures for association rules. (Lift)	F
9.	<u>Sorting a student database</u> based on student identification numbers is a data mining task. (database query)	F
10.	The goal of clustering analysis is to maximize the number of clusters . (maximize the intra-cluster similarity and minimize the inter-cluster similarity)	F
11.	The probability that two items are <u>bought together</u> is known as confidence . (support)	F
12.	The silhouette coefficient is a method to determine the natural number of clusters for hierarchical algorithms density-based algorithm (partitioning algorithms)	F
13.	All continuous variables are ratio. (Some)	F
14.	<u>Attributes</u> are sometimes called <u>variables</u> and <u>objects</u> are sometimes called <u>observations</u>	T
15.	Binary variables are sometimes continuous (always discrete)	F
16.	Computing the <i>total sales of a company</i> . Is a data mining task (accounting)	F
17.	Correlation analysis divides data into groups that are meaningful, useful, or both. (Cluster analysis)	F
18.	Database mining refers to the process of deriving high-quality information from text (Text mining)	F
19.	Dissimilarity matrix stores n data objects that have p attributes as an n-by-p matrix (Data)	F
20.	Data matrix stores a collection of proximities for all pairs of n objects as an <u>n-by-n</u> matrix (Dissimilarity)	F
21.	Dividing the customers of a company according to their profitability. Is a data mining task (accounting)	F
22.	For an association rule, if we move one item from the right-hand-side to the left-hand-side of the rule, then the confidence will never change. (support)	F
23.	If <u>all the proper subsets</u> of an itemset are frequent, then the itemset itself <u>must also be frequent</u> . (may)	F
24.	In decision tree algorithms, <u>attribute selection measures</u> are used to <u>rank attributes</u>	T
25.	In decision tree algorithms, attribute selection measures are used to <u>reduce the dimensionality</u> .	T
26.	In lazy learner we interest in the largest distance (minimum)	F
27.	Multimedia Mining is the application of data mining techniques to discover patterns <u>from the Web</u> . (Web data mining)	F
28.	Regression is a method of integration (correlation)	F
29.	Strategies for data transformation include chi-square test (integration)	F
30.	The Pruning make the decision tree more complex (reliable)	F
31.	An object is an outlier if its density is equal to the density of its neighbors. (significantly lower than)	F
32.	A common weakness of association rule mining is that it is not produce enough interesting rules (it produces too many rules)	F
33.	Accuracy is interestingness measures for association rules (correlation, lift)	F
34.	Core object is an object whose ϵ -neighbourhood contains objects less than MinPts (more than)	F



35.	Correlation analysis is used to <u>eliminate misleading rules</u>	T
36.	Correlation is a method of cleaning (analysis)	F
37.	Extracting the frequencies of a sound wave. Is a data mining task. (signal processing)	F
38.	Incomplete data problem can be solved by <i>binning</i> (binning is a smoothing technique)	F
39.	K-Nearest Neighbor <u>Classifiers</u> <u>do classification</u> when <i>new test data is available</i>	T
40.	Median is a value that occurs most frequently in the attribute values (Mode)	F
41.	Mode is a <i>middle value</i> in set of ordered values (Median)	F
42.	One strength of a <u>Bayesian Classifier</u> is that it <i>can be easily trained</i>	T
43.	Outlier analysis is a method of transformation (cleaning)	F
44.	Predicting the outcomes of tossing a (fair) pair of dice. Is a data mining Task? (probabilistic) (If the die were not fair this could be data mining task)	F
45.	Redundancy is an important issue in data cleaning (integration)	F
46.	Sampling methods smooth noisy data (reduction)	F
47.	The bottleneck of the Apriori algorithm caused by the number of association rules . (number of candidates and scans)	F
48.	the object is local outlier if it is deviate significantly from the rest of the dataset (global)	F
49.	One strength of a <u>Bayesian classifier</u> is that <i>can be easily trained</i>	T
50.	Mode is found in the numeric attribute value only (all types of attribute values)	F
51.	K-means algorithm is used for classification (clustering)	F
52.	K-nearest neighbor is same as k-means. (not the same)	F
53.	To measure objects similarity for objects with nominal attributes, use Euclidean distance. (numeric)	F
54.	Euclidean distance is used to measure dissimilarity of nominal attributes (numeric)	F
55.	RANDOMLY allocate 2/3 of data for training and remaining 1/3 for testing, called random subsampling . (Holdout)	F
56.	You must find the 5 number summary in order to make a box and whisker plot	T
57.	An OLAP system focuses mainly on the <u>current data within an enterprise</u> (OLTP)	F
58.	A data cube allows data to be modeled and viewed in single dimension (multiple dimensions)	F
59.	A data warehouse is a object-oriented, isolated, one variant, and volatile collection of data. (subject-oriented, integrated, time variant, nonvolatile)	F
60.	An OLAP system is customer-oriented (OLTP)	F
61.	OLTP is used to decision support (OLAP)	F
62.	In Star schema, a fact table is located in the middle connected to a set of dimension tables	T
63.	The access patterns of an OLAP system consist mainly of short and atomic transactions (OLTP)	F

